

PRIMITIVE GASSMANN TWINS FROM THE 27 LINES OF A CUBIC SURFACE

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ABSTRACT. We construct two explicit degree-320 number fields from the labelled 27 lines on one cubic surface: after scaling each line-eliminant root d_i to the algebraic integer $\alpha_i = Ld_i$, the fields are generated by quadratic orbit sums supported on respectively 27+27 and 81 unordered pairs. The known classification of Gassmann collisions in $W(E_6)$ identifies the underlying pair abstractly, but it does not provide primitive generators in the line coordinates or arithmetic that remains valid under the unresolved 3-adic decomposition-group choice. We prove primitivity by exact support stabilizers and 320 distinct reductions at the completely split prime 692717, and we combine the complete 350-class subgroup-character table with double-coset calculations for both permitted decomposition groups. The resulting nonisomorphic fields have the same Galois normal closure, equal Dedekind zeta functions, and realize the unique minimum-index collision within that complete $W(E_6)$ subgroup table. Both fields have signature $(16, 152)$, exact finite ramified support consisting of eight primes, and the same exact factorized positive discriminant. Nevertheless, their finite étale $\mathbf{Q}_3$ -algebras are nonisomorphic for each of the two retained decomposition-group alternatives; no alternative is selected.

1. INTRODUCTION AND THE INTEGRATED THEOREM

Equal Dedekind zeta functions conceal information at ramified primes. They force the same unramified residue-degree data, yet they need not determine the products of local fields at the excluded primes. The phenomenon is classical; the problem addressed here is to realize it primitively and exactly inside the line field of one cubic surface, while retaining every decomposition-group branch left by the preceding ramification analysis.

Let $Y/\mathbf{Q}$ be the fixed smooth cubic surface, let E be its degree-27 line field, and let K be its fixed normal closure. The released labelled line action identifies

$$G = \text{Gal}(K/\mathbf{Q}) = W(E_6), \quad |G| = 51840. \quad (1.1)$$

The complete subgroup computation of James already records 350 subgroup classes, 339 permutation-character profiles, and eleven Gassmann collisions [4, §3.1, p. 22]. An abstract collision K^{H+}, K^{H-} , however, does not supply primitive elements in the coordinates of the 27 lines. Nor does it determine whether the same local separator works for both decomposition groups at 3 that survive the ramification theorem for K .

We now give the durable data needed to state the result. In the fixed 27-point labelling, define H_+ and H_- by the following two displays:

$$\begin{aligned} H_+ = \langle & (3, 19, 6)(4, 21, 17)(5, 20, 16)(7, 24, 27)(8, 11, 23)(12, 15, 22), \\ & (1, 16, 23, 10, 7, 15)(2, 27, 11, 14, 5, 22)(3, 13, 20, 19, 9, 24) \\ & (4, 12, 18, 17, 8, 25)(6, 26, 21), \\ & (1, 26, 10)(2, 13, 25)(3, 22, 8)(4, 20, 27)(5, 24, 17)(6, 15, 23) \\ & (7, 21, 16)(9, 14, 18)(11, 19, 12) \rangle. \end{aligned} \quad (1.2)$$

$$\begin{aligned}
H_- = & \langle (1, 5, 3, 6, 4, 2)(7, 10, 17, 18, 20, 13)(8, 21, 16, 15, 9, 14) \\
& (11, 19, 12)(22, 26, 24, 27, 25, 23), \\
& (1, 7, 16, 11, 22, 10)(2, 15, 27)(3, 13, 9, 20, 24, 19) \\
& (4, 12, 8, 18, 25, 17)(5, 26, 23, 21, 14, 6), \\
& (1, 16, 22)(2, 23, 21)(3, 9, 24)(4, 8, 25)(5, 26, 15)(6, 27, 14) \rangle.
\end{aligned} \tag{1.3}$$

These arrays define the embedded groups. Their package positions 301 and 303 will be used only as frozen-version locators. For the two retained C58 decomposition-group alternatives at 3, ToM 140 and ToM 206 likewise denote versioned Table-of-Marks locators; the exact embedded arrays and filtration data, not these numbers, are the durable definitions.

Write the nonmonic degree-27 line eliminant as $g(d) \in \mathbf{Z}[d]$, let $L > 0$ be its leading coefficient, and label its roots $d_1, \dots, d_{27}$ as above. Set

$$\alpha_i = Ld_i, \quad \mathcal{S}_+ = H_+ \cdot \{1, 2\} \sqcup H_+ \cdot \{1, 9\}, \quad \mathcal{S}_- = H_- \cdot \{1, 2\}, \tag{1.4}$$

where the two H_+ -orbits are disjoint and have size 27 each, while $|\mathcal{S}_-| = 81$. Our generators are always the scaled integral sums

$$\eta_{\pm} = \sum_{\{i,j\} \in \mathcal{S}_{\pm}} \alpha_i \alpha_j. \tag{1.5}$$

If an unscaled sum is mentioned, it is denoted $\tilde{\eta}_{\pm}$ and $\eta_{\pm} = L^2 \tilde{\eta}_{\pm}$. The symbol $\eta_{\pm}$ never denotes the unscaled object.

Theorem 1.1 (Primitive $W(E_6)$ Gassmann twins). *With the exact embedded subgroups in (1.2) and (1.3) and the scaled sums in (1.5), put*

$$R_{\pm}(T) = \prod_{gH_{\pm} \in G/H_{\pm}} (T - g\eta_{\pm}), \quad F_{\pm} = \mathbf{Q}(\eta_{\pm}).$$

Then the following five assertions hold as one integrated statement.

- (i) *The polynomials $R_{\pm}(T)$ are monic integral irreducible separable polynomials of degree 320, in exact product form, and*

$$F_+ = K^{H_+}, \quad F_- = K^{H_-}.$$

- (ii) *Both subgroups have order 162, index 320, trivial core, and normalizer order 324. Their SmallGroup identifiers are [162, 11] and [162, 19], respectively. Their full rational transitive permutation characters agree, and $\{H_+, H_-\}$ is the unique minimum-index collision within the complete 350-class $W(E_6)$ subgroup table.*

- (iii) *The fields are not isomorphic over $\mathbf{Q}$, have common normal closure K , and satisfy*

$$\zeta_{F_+}(s) = \zeta_{F_-}(s).$$

- (iv) *Let q be the certified prime*

$$q = 14932047182473291995860108491583652133938007263719$$

and put

$$\begin{aligned}
A &= 181 \cdot 997 \cdot 2346241 = 423395612137, \\
B &= 283 \cdot 1801 \cdot q \\
&= 7610610604104534884325967676315830570581925356194091077.
\end{aligned}$$

Both fields have signature (16, 152), exact finite ramified support

$$\{3, 5, 181, 283, 997, 1801, 2346241, q\},$$

and signed field discriminant

$$\text{Disc}(F_+) = \text{Disc}(F_-) = +3^{624} 5^{496} A^{192} B^{160}.$$

- (v) For each retained alternative $D_3 = \text{ToM } 140$ and $D_3 = \text{ToM } 206$, the finite étale $\mathbf{Q}_3$ -algebras

$$F_+ \otimes_{\mathbf{Q}} \mathbf{Q}_3 \quad \text{and} \quad F_- \otimes_{\mathbf{Q}} \mathbf{Q}_3$$

are nonisomorphic. Hence F_+ and F_- are arithmetically equivalent but not locally equivalent, and their adele rings are nonisomorphic.

TABLE 1. The two primitive subfields have the same degree, normal closure, Dedekind zeta function, signature, and signed field discriminant. Their factor-degree multisets over $\mathbf{Q}_3$ differ in both retained decomposition-group alternatives. Here $A = 181 \cdot 997 \cdot 2346241$ and $B = 283 \cdot 1801 \cdot 14932047182473291995860108491583652133938007263719$, with $q = 14932047182473291995860108491583652133938007263719$ the certified prime; superscripts in the final two rows are multiplicities. Neither branch is selected. The displayed degree differences are sufficient product-algebra separators and do not classify the individual completion fields.

Datum	F_+	F_-	Logical role
<i>Explicit constructions</i>			
Support defining $\eta_{\pm}$	$27 + 27 = 54$ pairs	81 pairs	Distinct scaled integral orbit sums.
Stabilizer	$\text{SmallGroup}[162, 11],$ $H_+^{\text{ab}} \cong [2, 3]$	$\text{SmallGroup}[162, 19],$ $H_-^{\text{ab}} \cong [2]$	Nonconjugate order-162 stabilizers.
<i>Common global data</i>			
Degree / normal closure	320 / K	320 / K	Common explicit ambient normal closure.
Dedekind zeta function	$\zeta_{F_+}(s)$	$\zeta_{F_-}(s) = \zeta_{F_+}(s)$	Equality of full induced trivial characters.
Signature	(16, 152)	(16, 152)	Same archimedean data.
Signed field discriminant	$+3^{624}5^{496}A^{192}B^{160}$	$+3^{624}5^{496}A^{192}B^{160}$	Same conductor-derived global discriminant.
<i>Local separators at 3</i>			
$D_3 = \text{ToM } 140$: factor degrees	$1^8, 6^{10},$ $9^8, 18^{10}$	$2^4, 3^{12}, 6^4,$ $9^4, 18^{12}$	8 degree-one factors versus 0.
$D_3 = \text{ToM } 206$: factor degrees	$2^4, 12^5,$ $18^4, 36^5$	$4^2, 6^6, 12^2,$ $18^2, 36^6$	4 unramified quadratic factors versus 0 degree-two factors.

NO_BAD_EULER_OR_ROOT_NUMBER. No decomposition Frobenius, bad Artin Euler polynomial or factor, epsilon factor, root number, Artin-holomorphy statement, or automorphy statement is asserted. Here “primitive” means only that $\eta_{\pm}$ generate $K^{H_{\pm}}$; it does not assert that these extensions have no intermediate fields or that the coset actions are primitive.

The theorem joins four instance-specific contributions.

- (1) Exact integral quadratic sums in the labelled line roots generate both fixed fields; the proof uses support stabilizers and 320 distinct values at the split prime 692717.
- (2) The complete subgroup table identifies the table-restricted minimum-index collision, while trivial cores, nonconjugacy, and Artin formalism give common normal closure, field nonisomorphism, and equal zeta functions.

- (3) The inherited ramification package—complex conjugation together with the lower filtrations—yields signature $(16, 152)$, exact eight-prime support, and the signed discriminant in [theorem 1.1\(iv\)](#).
- (4) Separate complete double-coset tables prove local-algebra nonisomorphism for both D_3 alternatives; neither alternative is selected.

[Section 2](#) fixes the prior-work boundary and frozen data. [Section 3](#) proves integrality and primitivity. [Section 4](#) proves the complete-table and field-theoretic claims. [Section 5](#) derives the global arithmetic, and [section 6](#) treats the two local branches separately. [Sections 7](#) and [8](#) record the finite evidence boundary and the theorem’s limitations. The appendices give exact arrays, audit ledgers, and source locators.

2. PRIOR WORK, FROZEN INPUT, AND DURABLE DEFINITIONS

2.1. Arithmetic equivalence and its local boundary. Gassmann’s character condition and Perlis’s number-field formulation identify equality of Dedekind zeta functions with equality of the relevant induced trivial characters [[9](#), Theorem 1]. This criterion is the universal bridge used below; the present work does not propose a new criterion for arithmetic equivalence. Bosma and de Smit explain the common global invariants forced by arithmetic equivalence and use permutation characters computationally [[1](#), printed pp. 67–68]. Our contribution is narrower: we give primitive line-coordinate generators and exact ramified arithmetic for one fixed normal closure.

Equal zeta functions do not imply equality of all arithmetic attached to the ramified completions. Explicit common-normal-closure constructions with different ramified local degrees occur in the literature [[7](#), proof of Theorem 1.2], and explicit polynomial pairs can have identical zeta data but different ramification tuples [[6](#), Theorems 1.3 and 3.7]. Komatsu separates arithmetic equivalence from equivalence of adèle rings [[5](#)]. These works preclude any claim that local inequivalence or nonisomorphic adèle rings are new in general. What is specific here is that the same local obstruction is proved for each of the two decomposition groups still possible in this cubic-surface instance.

We call two number fields *locally equivalent* if their places can be put in a bijection under which paired completions are isomorphic as topological fields. Equivalently, their adèle rings are isomorphic as topological rings; we use this standard equivalence in the sense discussed by Komatsu [[5](#)].

2.2. Subgroup collisions and resolvents. James’s complete $W(E_6)$ computation is the prior source for the figures 350, 339, and eleven collision pairs [[4](#), §3.1, p. 22]. We neither rediscover those collisions nor call the pair in [theorem 1.1](#) a new Gassmann pair. The phrase “unique minimum-index” will always mean unique among those eleven collision buckets in the complete 350-class table. It is not a minimum-degree assertion about arithmetically equivalent fields in general.

The resolvent method is likewise classical. Stauduhar constructs resolvents from ordered roots and subgroup invariants [[10](#)]; Fieker and Klüners develop practical relative invariants and stabilizer methods [[3](#), §§1–2 and Remark 2.1]. Our orbit products use this established framework. Their special feature is the exact quadratic support in the labelled 27-line carrier and a split-prime noncollision proof of primitivity, not a new resolvent algorithm.

Explicit realization of subgroups of $W(E_6)$ through lines on cubic surfaces is also established. Elsenhans and Jahnke give a moduli and inverse Galois construction, including an algorithm that produces cubic surfaces with prescribed line action [[2](#), Theorem in the introduction, Algorithm 5.1, and Algorithm A.10]. Here that geometric program is an input boundary. The paper studies two particular primitive subfields and their global and local arithmetic inside one already fixed realization.

The source audit authorizes only the following bounded conjunction: for this one cubic-surface line extension, exact integral quadratic orbit sums primitively realize the table-restricted minimum-index $W(E_6)$ Gassmann pair, and the resulting fields have an exact eight-prime global arithmetic package and nonisomorphic finite étale $\mathbf{Q}_3$ -algebras in both surviving decomposition-group branches. This is a source boundary, not an absolute-priority or “first” claim.

2.3. The frozen cubic-surface input. Two released predecessor results are used without being re-proved. The fixed surface is $Y = V(\Phi) \subset \mathbf{P}_{\mathbf{Q}}^3$, where

$$\begin{aligned} \Phi = & 75081586157u_0^3 - 28576620789u_0^2u_1 - 122000922135u_0^2u_2 \\ & - 5364921951u_0^2u_3 + 164150208636u_0u_1^2 - 415458334296u_0u_1u_2 \\ & + 151070718312u_0u_1u_3 + 1158143874300u_0u_2^2 + 114691988016u_0u_2u_3 \\ & + 113572676646u_0u_3^2 + 6898957820u_1^3 + 1132596902196u_1^2u_2 \\ & - 30413540316u_1^2u_3 - 2054867641020u_1u_2^2 + 151980984216u_1u_2u_3 \\ & + 36794420832u_1u_3^2 + 2646295985484u_2^3 + 560186573940u_2^2u_3 \\ & + 706181383584u_2u_3^2 + 1884468968u_3^3. \end{aligned}$$

The line-field input supplies the nonmonic eliminant $g(d)$, its leading coefficient L , the reconstructed 27 lines, and the faithful action (1.1). These are electronic exact data rather than an implicit choice: the C56 certificate and release commit in table 7 define Φ at `payload.surface.primitive_coefficients`, define g at `payload.irreducibility.eliminant_coefficients_d_0_to_27` (with L its d^{27} coefficient), and give the four symbolic line equations at `payload.grassmann_main_chart.line_equations_sparse`. The C59 resolver evidence in the same table fixes the actual-to-standard label map and the arrays `payload.line_configuration.d_by_standard_label` and `payload.line_configuration.alpha_by_standard_label`. The characteristic-zero symbols d_i and $\alpha_i = Ld_i$ remain the labelled roots and scaled roots above. At the split prime $p = 692717$, the two resolver arrays store their reductions at zero-based JSON index $i - 1$; they do not store the characteristic-zero algebraic numbers themselves. Together with the actual-to-standard map, these entries make “labelled root” a recorded rather than implicit ordering convention.

The ramification input supplies the exact eight-prime support of K , the lower filtrations at 3 and 5, the tame inertia classes, complex conjugation, and the exhaustive alternative

$$D_3 = \text{ToM } 140 \quad \text{or} \quad D_3 = \text{ToM } 206, \quad I_3 = \text{ToM } 140 \text{ in either case.} \quad (2.1)$$

The second case has $[D_3 : I_3] = 2$. The analysis below carries both entries of (2.1) through separate tables and selects neither.

2.4. Durable subgroup data. Equations (1.2) and (1.3) are the field-defining subgroup data. For an alternative exact encoding, if a permutation is stored by its images of $1, \dots, 27$, the three generators of H_+ are

$$\begin{aligned} & (1, 2, 19, 21, 20, 3, 24, 11, 9, 10, 23, 15, 13, 14, 22, 5, 4, 18, 6, 16, 17, 12, 8, 27, 25, 26, 7), \\ & (16, 27, 13, 12, 22, 26, 15, 25, 24, 7, 14, 18, 20, 5, 1, 23, 8, 17, 9, 19, 6, 2, 10, 3, 4, 21, 11), \\ & (26, 13, 22, 20, 24, 15, 21, 3, 14, 1, 19, 11, 25, 18, 23, 7, 5, 9, 12, 27, 16, 8, 6, 17, 2, 10, 4), \end{aligned}$$

and those of H_- are

$$\begin{aligned} & (5, 1, 6, 2, 3, 4, 10, 21, 14, 17, 19, 11, 7, 8, 9, 15, 18, 20, 12, 13, 16, 26, 22, 27, 23, 24, 25), \\ & (7, 15, 13, 12, 26, 5, 16, 18, 20, 1, 22, 8, 9, 6, 27, 11, 4, 25, 3, 24, 14, 10, 21, 19, 17, 23, 2), \\ & (16, 23, 9, 8, 26, 27, 7, 25, 24, 10, 11, 12, 13, 6, 5, 22, 17, 18, 19, 20, 2, 1, 21, 3, 4, 15, 14). \end{aligned}$$

TABLE 2. Durable invariants of the two exact embedded subgroups. The final row consists only of version-sensitive locators in the frozen TomLib table; it is not a definition of either subgroup.

Invariant	H_+	H_-
Order / index	162/320	162/320
SmallGroup identifier	[162, 11]	[162, 19]
Abelianization invariants	[2, 3]	[2]
Derived subgroup order	27	81
Centre order	1	1
Core in G	1	1
Normalizer order in G	324	324
Frozen TomLib locator	301	303

This redundant exact encoding guards against cycle-convention ambiguity.

Lemma 2.1 (Embedded subgroup carrier). *The arrays above generate embedded subgroups of the released labelled action with the invariants in table 2. The pair supports in (1.4) have setwise stabilizers H_+ and H_- , respectively.*

Proof. The image arrays determine permutations without reference to a package ordering. Closure in the displayed 27-point action gives order 162 in each case, hence index $51840/162 = 320$. Exact enumeration gives the remaining group invariants in table 2. Acting on unordered pairs gives component sizes 27, 27 for H_+ and 81 for H_- ; the two H_+ components are disjoint. Enumerating the setwise stabilizer of the resulting 54- and 81-element supports returns exactly the displayed groups. The full pair arrays and their certificate hashes are recorded in appendix A; the finite enumeration is independently reconstructed as described in section 7. $\square$

The fixed version used for the finite table is GAP 4.11.1 with TomLib 1.2.9, whose table is named `U4(2).2`. GAP and TomLib supply computational conventions and versioned data, not the durable field definition [8, 11].

3. PRIMITIVE INTEGRAL QUADRATIC RESOLVENTS

3.1. Scaled roots and integrality. The leading coefficient is essential. Write

$$g(x) = Lx^{27} + b_{26}x^{26} + \cdots + b_1x + b_0, \quad b_i \in \mathbf{Z}. \quad (3.1)$$

Lemma 3.1 (Integral scaling). *Every $\alpha_i = Ld_i$ is an algebraic integer. Consequently $\eta_{\pm}$ are algebraic integers and $R_{\pm}(T) \in \mathbf{Z}[T]$.*

Proof. Substitute $x = y/L$ in (3.1) and multiply by L^{26} . The result is the monic integral equation

$$y^{27} + b_{26}y^{26} + Lb_{25}y^{25} + \cdots + L^{25}b_1y + L^{26}b_0 = 0.$$

It is satisfied by $y = Ld_i$, proving that α_i is integral. Products $\alpha_i\alpha_j$ and the finite sums (1.5) are therefore integral.

Because $H_{\pm}$ fix $\eta_{\pm}$, the products over $G/H_{\pm}$ are well-defined. Their coefficients are symmetric functions of a full G -orbit, hence lie in the fixed field $K^G = \mathbf{Q}$. They are also algebraic integers. A rational algebraic integer lies in $\mathbf{Z}$, so every coefficient of $R_{\pm}$ is integral. $\square$

Convention 3.2 (Scaled and unscaled sums). Throughout the paper, $\eta_{\pm}$ means exactly the scaled sum in (1.5). The optional unscaled quantity is

$$\tilde{\eta}_{\pm} = \sum_{\{i,j\} \in \mathcal{S}_{\pm}} d_i d_j, \quad \eta_{\pm} = L^2 \tilde{\eta}_{\pm}.$$

All integral-polynomial and modular-coefficient statements refer to $\eta_{\pm}$, never to $\tilde{\eta}_{\pm}$.

The finite product defining $R_{\pm}$ is an exact characteristic-zero object; an expanded coefficient vector is neither needed nor supplied. In particular, no characteristic-zero coefficient hash is asserted. This distinction matters because the hashes below bind modular products only.

3.2. Support stabilizers and formal orbits. The setwise support computation in lemma 2.1 gives

$$\text{Stab}_G(\mathcal{S}_+) = H_+, \quad \text{Stab}_G(\mathcal{S}_-) = H_-. \quad (3.2)$$

Thus each support has 320 translates. Equality (3.2) is a statement about formal supports. It does not by itself exclude an accidental equality between two evaluated quadratic sums. The next argument supplies exactly that missing bridge.

3.3. The split-prime noncollision argument. The prime used for the specialization is

$$p = 692717. \quad (3.3)$$

Exact arithmetic proves that p is prime, is outside the ramified support, and does not divide the eliminant or lexicographic reconstruction denominators. Modulo p , the normalized eliminant is squarefree and splits as 1^{27} . Hence all labelled roots and all scaled values can be reduced in one common residue field.

Lemma 3.3 (Split-prime noncollision). *For each sign, the 320 translated values $g\eta_{\pm}$, indexed by $gH_{\pm} \in G/H_{\pm}$, are pairwise distinct in characteristic zero. Consequently*

$$\text{Stab}_G(\eta_{\pm}) = H_{\pm}, \quad \mathbf{Q}(\eta_{\pm}) = K^{H_{\pm}}.$$

Proof. Reduce the labelled roots, their scaled coordinates, and the translated support sums modulo a fixed prime of the integral model above p . Complete enumeration gives 320 distinct residues for the H_+ family and 320 distinct residues for the H_- family. If two characteristic-zero conjugates were equal, their difference would be zero as an algebraic integer and would reduce to zero at every common integral reduction. Their residues would then coincide, a contradiction.

It follows that the G -orbit of $\eta_{\pm}$ has size 320. Since $H_{\pm}$ fixes $\eta_{\pm}$ and $[G : H_{\pm}] = 320$, orbit-stabilizer forces $\text{Stab}_G(\eta_{\pm}) = H_{\pm}$. Galois correspondence then gives

$$[\mathbf{Q}(\eta_{\pm}) : \mathbf{Q}] = 320, \quad \mathbf{Q}(\eta_{\pm}) = K^{H_{\pm}}.$$

□

As an independently checkable summary of the specialization, the monic degree-320 products modulo p have 321 coefficients and SHA-256 values

$$H_+: \quad 21b304679d3b77a7b1fae4182e203d8f2652588efffa4a160cccd98ac3e81257, \quad (3.4)$$

$$H_-: \quad 76fa8081c92e58839f60659fa7c9979d9b002fae5408cc30777341d21665acb2. \quad (3.5)$$

These are hashes of the complete modular coefficient arrays. They are not hashes of expanded characteristic-zero coefficients.

Corollary 3.4 (Exact product-form minimal polynomials). *The orbit products $R_{\pm}(T)$ are the monic integral irreducible separable minimal polynomials of $\eta_{\pm}$, each of degree 320.*

Proof. By lemma 3.1, the orbit products lie in $\mathbf{Z}[T]$. By lemma 3.3, each product contains every distinct $\mathbf{Q}$ -conjugate of $\eta_{\pm}$ exactly once. It is therefore the minimal polynomial. Characteristic zero gives separability. $\square$

3.4. Graph authority and labelling independence. The labelled action is tied back to the cubic surface rather than accepted as a free-standing abstract permutation representation. At the split prime, the exact lexicographic formulas reconstruct all 27 lines. Substitution into each of four chart line equations gives zero for every line. Pairwise incidence then yields a graph with 27 vertices, 135 edges, and degree 10 at every vertex. We denote this line-intersection graph by Γ_{meet} . It is the complement of the graph conventionally called the Schlaefli graph, whose edges join skew lines and which is therefore 16-regular; complementation does not change the automorphism group.

Lemma 3.5 (Full graph-labelling bridge). *Inside the fixed labelled action,*

$$\text{Aut}(\Gamma_{\text{meet}}) = G$$

as equality of the two complete 51,840-element permutation sets. Any other graph identification transports $H_{\pm}$ and $S_{\pm}$ by one element of G and merely permutes the 320 factors of $R_{\pm}$.

Proof. The six fixed simple-reflection generators preserve the reconstructed graph, so the released G is contained in the graph automorphism group. Exact enumeration gives $|G| = 51840$. Independently, individualization followed by equitable color refinement with base vertices 1, 2, 3, 4, 5 gives cell-choice sizes 27, 16, 10, 6, 2 and a discrete final partition. Their product is 51840, an upper bound for the graph automorphism group. Thus containment and the matching upper bound give equality of the complete permutation sets.

Two identifications of the reconstructed graph with the standard labelled graph differ by an automorphism, hence by some $w \in G$. They send $(H_{\pm}, S_{\pm})$ to $(wH_{\pm}w^{-1}, wS_{\pm})$. Left multiplication by w permutes the full coset orbit, leaving its product and noncollision property unchanged. Literal equality with the particular fixed fields in theorem 1.1 uses the displayed labelling; a relabelling transports the subgroups with it. $\square$

Eight sampled graph isomorphisms would not prove lemma 3.5. The argument uses all four chart equations, all 27 reconstructed lines, and equality of the full permutation sets. This completes the proof of theorem 1.1(i).

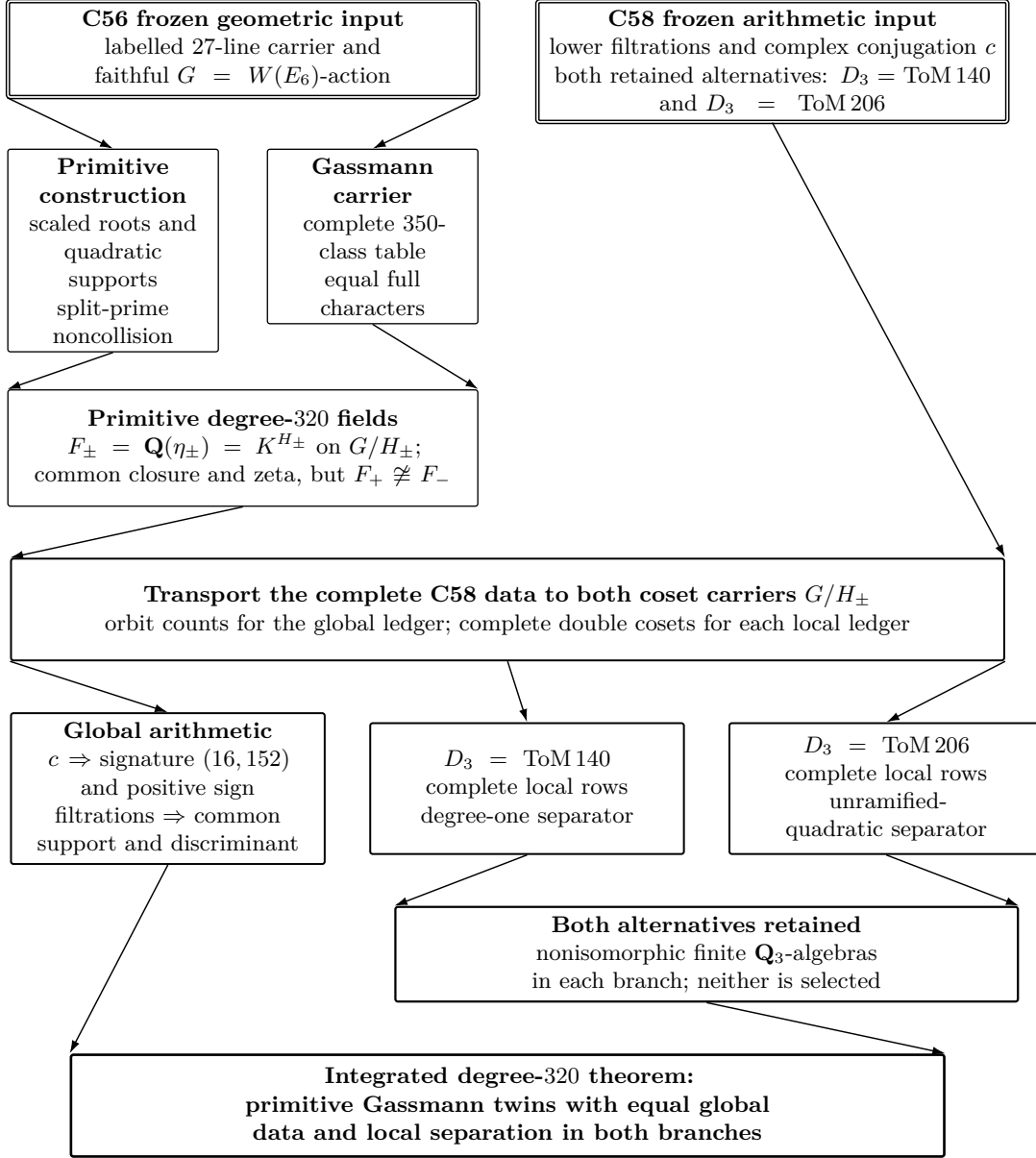


FIGURE 1. Proof dependencies for the integrated degree-320 statement. The frozen C56 labelled 27-line carrier is used separately for the primitive orbit-sum construction and the full-character Gassmann pair. The frozen C58 input consists of the lower filtrations, complex conjugation, and both exhaustive alternatives $D_3 = \text{ToM } 140$ and $D_3 = \text{ToM } 206$; these data are transported to the two coset carriers. The two local paths remain distinct: neither branch is selected, and each yields nonisomorphic finite $\mathbf{Q}_3$ -algebras.

4. THE COMPLETE GASSMANN TABLE AND FIELD CONSEQUENCES

4.1. **The complete collision table.** For a subgroup $H \leq G$, let

$$\chi_H = \text{Ind}_H^G \mathbf{1}$$

be the rational permutation character of G acting on G/H . The finite calculation compares the complete 25-entry character vector for each of all 350 subgroup conjugacy classes, not selected cyclic rows or sampled Frobenius data.

The prior count of 350 classes, 339 profiles, and eleven collisions is due to James [4, §3.1, p. 22]. The exact replay in the fixed U4(2) .2 table yields the complete projection in table 3.

TABLE 3. All collision buckets in the complete 350-class $W(E_6)$ subgroup table. The index is $[G : H]$ for either member of the bucket.

Bucket	Index	Bucket	Index
[12, 15]	12960	[17, 21]	12960
[29, 36]	8640	[31, 39]	8640
[41, 42]	6480	[46, 48]	6480
[57, 58]	6480	[59, 64]	6480
[112, 120]	3240	[132, 140]	2880
[301, 303]	320		

Proposition 4.1 (Complete-table minimum). *The equality $\chi_{H_+} = \chi_{H_-}$ holds on every conjugacy class of G . Among the eleven buckets in table 3, [301, 303] is the unique bucket of minimum index, namely 320.*

Proof. Grouping the 350 complete character vectors by equality gives precisely the eleven two-class buckets displayed in table 3; all other profiles occur once. The common vector for rows 301 and 303 is

$$(320, 0, 0, 32, 8, 20, 0, 0, 0, 0, 0, 0, 2, 0, 0, 16, 0, 0, 0, 0, 4, 0, 0, 0).$$

The index entries in table 3 show that 320 is strictly smaller than the other ten bucket indices. This proves uniqueness only within the complete $W(E_6)$ table. $\square$

The result is not inferred from a sample of good primes. The exact carrier contains all 350×25 character entries; its checksum-bound form and the independent regrouping check are described in appendix B.

4.2. Common normal closure and nonisomorphism.

Lemma 4.2 (Normal closure from the core). *For $F = K^H$, the normal closure of $F/\mathbf{Q}$ inside K is $K^{\text{Core}_G(H)}$. In particular, both F_+ and F_- have normal closure K .*

Proof. The conjugates of K^H inside K are $K^{gHg^{-1}}$. Their compositum is fixed by the intersection of their fixing groups:

$$\bigcap_{g \in G} gHg^{-1} = \text{Core}_G(H).$$

This compositum is the normal closure inside K . Both cores are trivial by table 2, so both normal closures equal K . $\square$

Lemma 4.3 (Common-normal-closure conjugacy criterion). *The fields F_+ and F_- are not isomorphic over $\mathbf{Q}$.*

Proof. Their subgroup abelianizations have invariant factors $[2, 3]$ and $[2]$. Thus H_+ and H_- are not isomorphic and, in particular, are not conjugate in G . This subgroup distinction alone is not yet the desired field statement.

Suppose instead that a $\mathbf{Q}$ -isomorphism $F_+ \rightarrow F_-$ existed. Extend it to an automorphism σ of $\overline{\mathbf{Q}}$. By lemma 4.2, both fields have the same normal closure K , so $\sigma(K) = K$. The restriction of σ to K belongs to G and transports point stabilizers:

$$\sigma H_+ \sigma^{-1} = H_-.$$

This contradicts nonconjugacy. Hence no such field isomorphism exists. $\square$

4.3. Artin formalism and the common zeta function. For any subgroup $H \leq G$, Artin formalism gives

$$\zeta_{K^H}(s) = L\left(s, \text{Ind}_H^G \mathbf{1}\right). \quad (4.1)$$

This is the classical Gassmann–Perlis bridge [9, Theorem 1]; permutation-character computations in arithmetic equivalence are also treated by Bosma and de Smit [1, printed p. 68].

Proposition 4.4 (Zeta equality). *The two Dedekind zeta functions agree:*

$$\zeta_{F_+}(s) = \zeta_{F_-}(s).$$

Proof. By the split-prime noncollision bridge in lemma 3.3, $F_{\pm} = K^{H_{\pm}}$. (The separate corollary 3.4 identifies the product-form minimal polynomials.) Substitute $H = H_+$ and $H = H_-$ in (4.1). The full character identity in proposition 4.1 makes the two Artin L -functions identical. $\square$

From the equality in proposition 4.4 alone we infer none of integral permutation equivalence, isomorphic rings of integers, equality of class numbers, local equivalence, adelic equivalence, or isomorphic idele groups. The local tables in section 6 separately disprove local and adelic equivalence for the present pair. Together, propositions 4.1 and 4.4 and lemmas 4.2 and 4.3 prove theorem 1.1(ii)–(iii).

5. SIGNATURE, CONDUCTORS, AND THE EXACT GLOBAL DISCRIMINANT

The global invariants are computed on the two 320-point coset carriers. The equality of permutation characters predicts equality of their conductor data, but the proof below uses the exact orbit counts supplied by the fixed lower filtrations.

5.1. Complex conjugation and the signature. Let $c \in G$ be the fixed complex-conjugation element. On each carrier $G/H_{\pm}$, the cyclic group $\langle c \rangle$ has 168 orbits.

Lemma 5.1 (Signature). *Both fields have signature $(r_1, r_2) = (16, 152)$, and their discriminants have positive sign.*

Proof. Fixed cosets correspond to real embeddings and two-element orbits correspond to conjugate pairs. Therefore

$$r_1 + 2r_2 = 320, \quad r_1 + r_2 = 168.$$

Subtracting gives $r_2 = 152$ and then $r_1 = 16$. The sign of a number-field discriminant is $(-1)^{r_2} = +1$. $\square$

5.2. Permutation conductors from orbit deficits. Let $V_H = \mathbf{Q}[G/H]$. If a subgroup $J \leq G$ has $o_H(J)$ orbits on G/H , then

$$\dim V_H^J = o_H(J), \quad \text{codim } V_H^J = 320 - o_H(J). \quad (5.1)$$

For a lower filtration $G_0 \supseteq G_1 \supseteq \cdots$ with inertia G_0 , the Artin conductor of the permutation representation is

$$a(V_H) = \sum_{u \geq 0} \frac{|G_u|}{|G_0|} (320 - o_H(G_u)). \quad (5.2)$$

The conductor–discriminant identity identifies this integer with the prime exponent of $\text{Disc}(K^H)$. This is a field-discriminant computation; it does not use a defining-polynomial discriminant.

The common exact orbit vector, in the order $(I_3, P_3, Q_3, I_5, P_5, C_3, C_2, C_\infty)$, is

$$(36, 56, 112, 16, 64, 128, 160, 168). \quad (5.3)$$

Here P_p denotes wild inertia, Q_3 the last nontrivial deep subgroup needed below, C_3 a tame cubic inertia class, and C_2 the reflection class.

Proposition 5.2 (Exact conductor exponents). *For either $H = H_+$ or $H = H_-$, the field K^H has common conductor exponents*

$$v_3 = 624, \quad v_5 = 496, \quad v_A = 192, \quad v_B = 160.$$

Proof. At 3, the lower filtration has inertia order 18, first wild group order 9, and six occurrences of the deep order-3 group. Substituting (5.3) in (5.2) gives

$$\begin{aligned} v_3 &= (320 - 36) + \frac{9}{18}(320 - 56) + 6\frac{3}{18}(320 - 112) \\ &= (320 - 36) + \frac{320 - 56}{2} + (320 - 112) \\ &= 284 + 132 + 208 = 624. \end{aligned}$$

At 5, inertia has order 20 and the order-5 wild group occurs in three positive lower grades. Hence

$$\begin{aligned} v_5 &= (320 - 16) + 3\frac{5}{20}(320 - 64) \\ &= (320 - 16) + \frac{3}{4}(320 - 64) \\ &= 304 + 192 = 496. \end{aligned}$$

At a tame cubic prime only $G_0 = C_3$ contributes, and at a reflection prime only $G_0 = C_2$ contributes. Formula (5.1) therefore gives

$$v_A = 320 - 128 = 192, \quad v_B = 320 - 160 = 160.$$

The orbit vector is identical on both coset carriers, so the calculation applies to both fields. $\square$

TABLE 4. Global arithmetic ledger. The coefficients in the conductor rows come from the complete lower filtrations, not from a polynomial discriminant.

Place/type	Orbit counts used	Field-discriminant contribution
3	(36, 56, 112)	$(320 - 36) + \frac{1}{2}(320 - 56) + (320 - 112) = 624$
5	(16, 64)	$(320 - 16) + \frac{3}{4}(320 - 64) = 496$
tame C_3 primes	128	$320 - 128 = 192$
reflection C_2 primes	160	$320 - 160 = 160$
infinity	168	$(r_1, r_2) = (16, 152)$; sign +

5.3. Exact support and signed identity. Let q denote the certified prime below, and define

$$q = 14932047182473291995860108491583652133938007263719, \quad (5.4)$$

$$A = 181 \cdot 997 \cdot 2346241 = 423395612137, \quad (5.5)$$

$$B = 283 \cdot 1801 \cdot q = 7610610604104534884325967676315830570581925356194091077. \quad (5.6)$$

Corollary 5.3 (Signed field discriminant and exact support). *For both signs,*

$$\text{Disc}(F_{\pm}) = +3^{624}5^{496}A^{192}B^{160},$$

and the finite ramified support is exactly

$$\{3, 5, 181, 283, 997, 1801, 2346241, q\}.$$

Proof. The normal closure K is unramified outside the displayed eight primes. Every subfield of K is therefore unramified outside the same set, giving one inclusion. Conversely, each exponent in [proposition 5.2](#) is positive. After expanding A and B , every one of the eight primes occurs with positive field-discriminant valuation, hence ramifies in each $F_{\pm}$. This gives the reverse inclusion. The positive sign is [lemma 5.1](#). $\square$

The positive discriminant is an 11,658-digit integer. As a reproducibility check, its unsigned no-newline decimal expansion has SHA-256

$$7f3ed0f731e5905f9af8254df2114ad15c2bb7d96cfa9a8b464a58ae8ea3ae70.$$

The decimal expansion itself is not needed in the paper. Neither the factorization nor its digest asserts an integral basis, a maximal order, monogenicity, or equality between the discriminant of an expanded degree-320 polynomial and the field discriminant. This completes the proof of [theorem 1.1\(iv\)](#).

6. BOTH 3-ADIC BRANCHES AND LOCAL SEPARATION

The two decomposition alternatives in [\(2.1\)](#) are now treated separately. We first give the common completion and lower-filtration formulas, including the bridge between computational right-coset orbits and the standard double-coset convention.

6.1. Double cosets and completion factors. Fix a prime w of K above 3, with decomposition group D and inertia group I . Let $F = K^H$. For a representative g of a double coset in $D \backslash G / H$, put

$$J = D \cap gHg^{-1}. \quad (6.1)$$

Lemma 6.1 (Completion factors). *The factors of the finite étale algebra $F \otimes_{\mathbf{Q}} \mathbf{Q}_3$ are indexed by $D \backslash G / H$. The factor attached to DgH has*

$$n = [D : J], \quad e = [I : I \cap J], \quad f = [D : IJ], \quad n = ef. \quad (6.2)$$

Proof. Embeddings of F into K correspond to cosets G/H . The decomposition group D acts on these embeddings, and its orbits are precisely the double cosets $D \backslash G/H$. One D -orbit therefore gives one prime of F above 3, equivalently one completion factor. The stabilizer in D of gH is $D \cap gHg^{-1} = J$, so the degree is $n = [D : J]$.

The inertia orbit inside that transitive factor has size $[I : I \cap J] = e$. Passing to the residue quotient gives $f = [D : IJ]$. Since $I \triangleleft D$, the product IJ is a subgroup and

$$[D : J] = [D : IJ][IJ : J] = [D : IJ][I : I \cap J] = fe.$$

Some computations enumerate right cosets and right orbits. Inversion sends $Hg^{-1}D$ to DgH and preserves stabilizer indices, so it yields the same completion factors rather than a second convention-dependent table. $\square$

6.2. Lower filtration and the local different entry. Let

$$D = G_{-1} \supseteq G_0 = I \supseteq G_1 \supseteq G_2 \supseteq \cdots$$

be the lower ramification filtration. The Artin conductor of the transitive permutation representation on D/J gives the discriminant contribution of the corresponding factor. A direct orbit calculation yields

$$fd = \sum_{u \geq 0} \frac{|G_u|}{|G_0|} \left([D : J] - [D : J] \frac{|G_u \cap J|}{|G_u|} \right). \quad (6.3)$$

Using (6.2), this is equivalent to the general formula

$$d = \sum_{u \geq 0} \frac{|G_u| - |G_u \cap J|}{|I \cap J|}. \quad (6.4)$$

For every representative in the two certified tables, the stored intersection quadruples in [appendix D](#) give

$$|J| = \frac{|D|}{n}, \quad |I \cap J| = \frac{|I|}{e}. \quad (6.5)$$

These two quantities are equal row by row. In the ToM 140 table this is immediate from $D = I$ and $n = e$. In the ToM 206 table, $|D| = 36$, $|I| = 18$, and every displayed row has $n = 2e$, so again $36/n = 18/e$. Since $I \cap J \leq J$ and the two finite groups have equal order, $J = I \cap J$ and therefore $J \leq I$. Thus the displayed entry used below is

$$d(g, H) = \sum_{u \geq 0} \frac{|G_u| - |G_u \cap J|}{|J|}. \quad (6.6)$$

Formula (6.6) explains, rather than merely labels, the fourth coordinate of each (n, e, f, d) row. Summing fd over all factors must reproduce the global exponent 624.

For this check, one must first reconcile the tables with the common Dedekind zeta function. The relevant datum is the multiset of residue degrees. In the ToM 140 branch, both fields have 36 factors of residue degree 1. In the ToM 206 branch, both have 18 factors of residue degree 2. Thus the different local-degree multisets below do not contradict [proposition 4.4](#). This is only a Dedekind-zeta residue-degree reconciliation; no bad Artin Euler factor is computed.

6.3. The retained branch $D_3 = \text{ToM 140}$. Here $D = I$ has order 18, with the exact chain

$$(D, I, P, Q) = (\text{ToM 140}, \text{ToM 140}, \text{ToM 72}, \text{ToM 7}), \quad (|D|, |I|, |P|, |Q|) = (18, 18, 9, 3). \quad (6.7)$$

The complete collected table is as follows.

TABLE 5. Complete collected local rows for the retained branch $D_3 = I_3 = \text{ToM } 140$. A term $(n, e, f, d)^m$ records m completion factors with degree $n = ef$, ramification index e , residue degree f , and local discriminant-exponent entry d . For each field the table independently gives the factor count, $\sum mn = 320$, and $\sum mfd = 624$. The 8 degree-one factors of F_+ , absent for F_- , suffice to separate the two finite $\mathbf{Q}_3$ -algebras; these rows do not classify the individual completion fields. This branch is retained alongside ToM 206, and neither branch is selected.

Field	Complete $(n, e, f, d)^{\text{multiplicity}}$ rows	Factors	$\sum mn$	$\sum mfd$
F_+	$(1, 1, 1, 0)^8, (6, 6, 1, 11)^{10},$ $(9, 9, 1, 18)^8, (18, 18, 1, 37)^{10}$	36	320	624
F_-	$(2, 2, 1, 1)^4, (3, 3, 1, 5)^{12}, (6, 6, 1, 11)^4,$ $(9, 9, 1, 18)^4, (18, 18, 1, 37)^{12}$	36	320	624

For F_+ , the degree and different checks read

$$8 \cdot 1 + 10 \cdot 6 + 8 \cdot 9 + 10 \cdot 18 = 320,$$

$$8(1)(0) + 10(1)(11) + 8(1)(18) + 10(1)(37) = 624.$$

For F_- they read

$$4 \cdot 2 + 12 \cdot 3 + 4 \cdot 6 + 4 \cdot 9 + 12 \cdot 18 = 320,$$

$$4(1)(1) + 12(1)(5) + 4(1)(11) + 4(1)(18) + 12(1)(37) = 624.$$

The multiplicities sum to 36 in either line, and every row satisfies $n = ef$. The product for F_+ has eight degree-one factors; the product for F_- has none.

6.4. The retained branch $D_3 = \text{ToM } 206$. Here D has order 36 and contains a unique normal embedded inertia group $I = \text{ToM } 140$, with chain

$$(D, I, P, Q) = (\text{ToM } 206, \text{ToM } 140, \text{ToM } 72, \text{ToM } 7), \quad (|D|, |I|, |P|, |Q|) = (36, 18, 9, 3). \quad (6.8)$$

The double-coset enumeration for this branch is independent of the ToM 140 branch table, although it uses the shared certified inertia chain $I = \text{ToM } 140 \supset \text{ToM } 72 \supset \text{ToM } 7$.

TABLE 6. Complete collected local rows for the retained branch $D_3 = \text{ToM } 206$, with its unique normal embedded inertia group $I_3 = \text{ToM } 140$. A term $(n, e, f, d)^m$ records m completion factors with degree $n = ef$, ramification index e , residue degree f , and local discriminant-exponent entry d . For each field the table independently gives the factor count, $\sum mn = 320$, and $\sum mfd = 624$. The 4 unramified quadratic factors of F_+ , while F_- has no degree-two factor, suffice to separate the finite $\mathbf{Q}_3$ -algebras; these rows do not classify the individual completion fields. This branch is retained alongside ToM 140, and neither branch is selected.

Field	Complete $(n, e, f, d)^{\text{multiplicity}}$ rows	Factors	$\sum mn$	$\sum mfd$
F_+	$(2, 1, 2, 0)^4$, $(12, 6, 2, 11)^5$, $(18, 9, 2, 18)^4$, $(36, 18, 2, 37)^5$	18	320	624
F_-	$(4, 2, 2, 1)^2$, $(6, 3, 2, 5)^6$, $(12, 6, 2, 11)^2$, $(18, 9, 2, 18)^2$, $(36, 18, 2, 37)^6$	18	320	624

For F_+ the independent checks are

$$4 \cdot 2 + 5 \cdot 12 + 4 \cdot 18 + 5 \cdot 36 = 320,$$

$$4(2)(0) + 5(2)(11) + 4(2)(18) + 5(2)(37) = 624.$$

For F_- they are

$$2 \cdot 4 + 6 \cdot 6 + 2 \cdot 12 + 2 \cdot 18 + 6 \cdot 36 = 320,$$

$$2(2)(1) + 6(2)(5) + 2(2)(11) + 2(2)(18) + 6(2)(37) = 624.$$

Each multiplicity sum is 18 and every row again has $n = ef$. The four $(2, 1, 2, 0)$ factors for F_+ are the unramified quadratic extension of $\mathbf{Q}_3$, repeated four times. No factor of F_- has degree 2.

6.5. One-way separation and branch-independent conclusion.

Proposition 6.2 (Complete branchwise local tables). *The tables in tables 5 and 6 contain every factor for the respective branch. In each row $n = ef$; in each field and branch the degrees sum to 320 and the values fd sum to 624. The residue-degree counts agree between F_+ and F_- branchwise.*

Proof. The D -orbits on $G/H_{\pm}$ exhaust the relevant coset carriers. Exact stabilizer intersections give (6.2), and the complete lower filtration gives (6.6). Collecting identical rows produces exactly the displayed multiplicities. The explicit sums above verify the three independent totals. Finally, all rows in the first branch have $f = 1$ and there are 36 of them, while all rows in the second have $f = 2$ and there are 18 of them. $\square$

Corollary 6.3 (Local and adelic inequivalence). *For either choice in (2.1),*

$$F_+ \otimes_{\mathbf{Q}} \mathbf{Q}_3 \not\cong F_- \otimes_{\mathbf{Q}} \mathbf{Q}_3.$$

Consequently the fields are not locally equivalent and their adele rings are not isomorphic.

Proof. An isomorphism between finite products of finite extensions of $\mathbf{Q}_3$ permutes their field factors and preserves the factor-degree multiset. In the ToM 140 branch, eight degree-one factors occur only for F_+ . In the ToM 206 branch, four degree-two factors occur only for F_+ . Each difference is therefore sufficient for product-algebra nonisomorphism. A single nonisomorphic local product rules out local equivalence. Moreover, an isomorphism of adele rings induces a bijection of their local-field factors; residue characteristic is preserved, so it would in particular identify the two products above 3. Their nonisomorphism therefore also rules out an isomorphism of adele rings. $\square$

The argument uses the tuples (n, e, f, d) only in the sufficient direction. Such a tuple does not classify an individual high-degree local field, and no classification of the completion factors is claimed. Both branches have been proved and neither has been selected. This completes [theorem 1.1\(v\)](#).

7. EXACT COMPUTATION AND THE REPRODUCIBILITY BOUNDARY

The proof separates finite premises from mathematical implications. Exact computation supplies embedded arrays, complete enumerations, and arithmetic identities. The preceding sections supply the bridges from those finite facts to integrality, fixed fields, normal closures, zeta functions, discriminants, and local products. A machine Boolean is never substituted for one of those implications.

7.1. The G0–G7 evidence map. The finite package is organized into eight gates.

- G0:** Rebind the complete released C56 and C58 authority, including their inventories, manifests, certificates, line carrier, and lower-filtration arrays.
- G1:** Reconstruct the primitive integral orbit sums, all 27 lines and four chart equations, the full Schlaefli automorphism set, and the split-prime noncollision products.
- G2:** Enumerate the complete 350-class table, all full permutation characters, the eleven buckets, and the durable subgroup invariants.
- G3:** Bind the exact premises used by the fixed-field, common-normal- closure, nonisomorphism, and Artin-formalism arguments.
- G4:** Recompute the common orbit vector, signature, sign, conductor exponents, support, and discriminant digest.
- G5:** Enumerate every double coset and row in the ToM 140 branch.
- G6:** Prove the unique normal ToM 140 inertia inside ToM 206 and enumerate the second complete local table without selecting a branch.
- G7:** Enforce checker independence, exact schemas, mutation and source rebound, scope leaves, and nonrelease status.

The producer and checker have disjoint theorem call graphs. Common code is limited to canonical input/output, fingerprints, and backend preflight. The checker rebuilds the complete scalar-leaf inventory, enforces exact key sets and strict types, and compares independently generated projections of the group and resolvent carriers. Historical bounded pilots were used to select the target, but they are not theorem authority and are not substituted for the promoted G1 reconstruction.

7.2. Finite audit totals. The frozen code/results tuple contains 13 source files and 8 result files, hence 21 live leaves. Its self-excluding scoped manifest binds 20 entries. The certificate payload has 15 top-level keys and 10,412 scalar leaves. The independent checker rejected 20,894 certificate mutations and 8 self-consistent evidence-carrier rebound mutations. All 48 source tests passed. A mandatory nonmutating replay and an independent post-refresh audit read the same promoted bytes.

These counts certify a finite reconstruction envelope. They do not prove that the present prose is correct, that a TeX build succeeds, that a paper has passed an independent manuscript audit, or that C59 has been promoted or released. The principal artifact identities and the exact inventories appear in [appendix E](#).

7.3. Software and conventions. The group lane uses GAP 4.11.1, TomLib 1.2.9, Small-Grp 1.4.1, and CTbLib 1.3.1. These package and version strings are reproducibility metadata. The mathematical Python lane uses Python 3.12.3, python-flint 0.9.0, SymPy 1.14.0, and NetworkX 3.5. The table-of-marks locators in this paper are meaningful only relative to that frozen version; the exact 27-point arrays remain authoritative. The specific software and table conventions used in the proof are those documented by official GAP and TomLib sources [8, 11].

The literal firewall in [theorem 1.1](#) is part of the certificate as well: a finite PASS creates no excluded analytic datum and constitutes neither paper nor release approval.

8. SCOPE, LIMITATIONS, AND CONCLUSION

8.1. Limitations and interpretation. The decomposition group at 3 remains unresolved. This is not hidden by a choice of notation: [table 5](#) and [table 6](#) are logically separate inputs, and the theorem was designed to be valid in both cases. Resolving the branch would be a different problem and is not necessary for the local separator proved here.

The exact minimal polynomials are supplied as orbit products, which define them completely over $\mathbf{Q}$. We neither print nor hash their expanded coefficient vectors. Likewise, the field discriminants follow from permutation conductors. We do not compute an integral basis or maximal order of either degree-320 field, assert monogenicity, or identify a polynomial discriminant with the field discriminant.

The local rows record degree, ramification index, residue degree, and a lower- filtration discriminant entry. Their differing factor-degree multisets separate the product algebras. They do not determine an individual high-degree completion up to isomorphism. From the zeta identity alone we infer none of integral permutation equivalence, isomorphic rings of integers, equal class numbers, local equivalence, adelic equivalence, or isomorphic idele groups. The local and adelic nonisomorphisms proved in [corollary 6.3](#) are separate positive conclusions.

Finally, the prior-work conclusion is deliberately bounded. Prior art includes James’s collision list, Gassmann–Perlis arithmetic equivalence, relative resolvents, and locally inequivalent arithmetically equivalent fields. The precise source and priority boundary is the one stated in [section 2](#) and [appendix F](#); no stronger priority claim is made here.

8.2. Conclusion. The labelled 27 lines turn a known abstract $W(E_6)$ collision into two primitive explicit subfields of one geometric normal closure. Exact support stabilizers and the split prime 692717 identify the fields; the complete subgroup table and Artin formalism give nonisomorphic fields with the same Dedekind zeta function; and the inherited ramification package—complex conjugation together with the lower filtrations—yields the common signature $(16, 152)$ and signed discriminant $+3^{624}5^{496}A^{192}B^{160}$. The equal global data conceal a different 3-adic product in both the ToM 140 and ToM 206 alternatives. The construction therefore requires no branch selection.

NO_BAD_EULER_OR_ROOT_NUMBER: no decomposition Frobenius, bad Artin Euler polynomial or factor, local epsilon factor, local or global root number, Artin holomorphy, or automorphy is claimed. We also make no rational-point, Hasse-principle, weak-approximation, Brauer–Manin, motive, Riemann-hypothesis, or Hilbert–Polya-operator claim. The machine and formal premises predate this manuscript; paper audit, compilation, archive, promotion, and release are separate downstream gates and are not asserted here.

APPENDIX A. SPLIT-PRIME AND GRAPH CERTIFICATE

This appendix records the finite objects used in the proofs of [lemmas 2.1, 3.3](#) and [3.5](#). All indices are one-based and refer to the fixed line labelling. Thus the lists below, together with the cycle arrays in [equations \(1.2\)](#) and [\(1.3\)](#), are independent of subgroup-table positions.

A.1. The two supports. The first orbit in $\mathcal{S}_+$ is

$$\begin{aligned}\mathcal{O}_{+,1} = \{ & \{1, 2\}, \{1, 18\}, \{2, 18\}, \{3, 6\}, \{3, 19\}, \{4, 5\}, \\ & \{4, 7\}, \{5, 7\}, \{6, 19\}, \{8, 11\}, \{8, 23\}, \{9, 13\}, \\ & \{9, 26\}, \{10, 14\}, \{10, 25\}, \{11, 23\}, \{12, 15\}, \{12, 22\}, \\ & \{13, 26\}, \{14, 25\}, \{15, 22\}, \{16, 17\}, \{16, 27\}, \{17, 27\}, \\ & \{20, 21\}, \{20, 24\}, \{21, 24\}\}.\end{aligned}$$

The second orbit is

$$\begin{aligned}\mathcal{O}_{+,2} = \{ & \{1, 9\}, \{1, 25\}, \{2, 14\}, \{2, 26\}, \{3, 12\}, \{3, 23\}, \\ & \{4, 16\}, \{4, 24\}, \{5, 21\}, \{5, 27\}, \{6, 11\}, \{6, 22\}, \\ & \{7, 17\}, \{7, 20\}, \{8, 15\}, \{8, 19\}, \{9, 25\}, \{10, 13\}, \\ & \{10, 18\}, \{11, 22\}, \{12, 23\}, \{13, 18\}, \{14, 26\}, \{15, 19\}, \\ & \{16, 24\}, \{17, 20\}, \{21, 27\}\}.\end{aligned}$$

Thus $\mathcal{S}_+ = \mathcal{O}_{+,1} \sqcup \mathcal{O}_{+,2}$ has 54 pairs. Its family-list, stabilizer-inventory, value-list, and sorted-value digests are respectively

```
0b2fa74280691fd83d364d150ff187431b548b8e9e8c951e6dbae034c189c2ec,
500f61f9101ca424c9ff142526c26e41fc2b06c1478b7f2feaf626f07c28aafe,
950b67aa2693ddf9608a22e90348c0a7dfdb71942611fa8ff4fbf392c649afa7,
289d4fcdd86554490a4208ea4fb8ead6abcd1e1d3b8b162c6e2d3805dd0fcee5.
```

The negative support is the following single 81-element orbit:

$$\begin{aligned}\mathcal{S}_- = \{ & \{1, 2\}, \{1, 5\}, \{1, 6\}, \{1, 13\}, \{1, 19\}, \{1, 20\}, \\ & \{2, 3\}, \{2, 4\}, \{2, 11\}, \{2, 18\}, \{2, 20\}, \{3, 5\}, \\ & \{3, 6\}, \{3, 7\}, \{3, 10\}, \{3, 11\}, \{4, 5\}, \{4, 6\}, \\ & \{4, 12\}, \{4, 17\}, \{4, 18\}, \{5, 7\}, \{5, 12\}, \{5, 13\}, \\ & \{6, 10\}, \{6, 17\}, \{6, 19\}, \{7, 9\}, \{7, 15\}, \{7, 24\}, \\ & \{7, 26\}, \{8, 12\}, \{8, 17\}, \{8, 18\}, \{8, 23\}, \{8, 26\}, \\ & \{8, 27\}, \{9, 10\}, \{9, 11\}, \{9, 23\}, \{9, 26\}, \{9, 27\}, \\ & \{10, 14\}, \{10, 24\}, \{10, 27\}, \{11, 21\}, \{11, 23\}, \{11, 24\}, \\ & \{12, 15\}, \{12, 25\}, \{12, 26\}, \{13, 15\}, \{13, 16\}, \{13, 22\}, \\ & \{13, 26\}, \{14, 17\}, \{14, 19\}, \{14, 22\}, \{14, 24\}, \{14, 25\}, \\ & \{15, 22\}, \{15, 24\}, \{15, 25\}, \{16, 19\}, \{16, 20\}, \{16, 23\}, \\ & \{16, 26\}, \{16, 27\}, \{17, 25\}, \{17, 27\}, \{18, 21\}, \{18, 23\}, \\ & \{18, 25\}, \{19, 22\}, \{19, 27\}, \{20, 21\}, \{20, 22\}, \{20, 23\}, \\ & \{21, 22\}, \{21, 24\}, \{21, 25\}\}.\end{aligned}$$

Its corresponding four digests are

```
4e329548ae15632deabfa8c1bea11b91d2fc7c60ddfc86ee4110d86a54d9cb66,
11b388fe39f5e884f064c38fda924cd14e9d063736cedee16d2b9cc01b3c1996,
aaffea3db8c0c913517562f702d07ab9f1b86c78ee90821d7aa09984f6d239c7,
b45cbd50b4a2c278416b6dd4a83ac7b160ad5113ff1b7d2a96e6c8961c0ee36b.
```

A.2. Finite-field noncollision. At the certified split prime $p = 692717$, the eliminant has 27 distinct roots in $\mathbf{F}_{692717}$. The ordered root vector has digest

```
96c072b9f1cc6a3e07d28433ed4b864b50f7d6841ed84a6536fd43ee050017dd.
```

For each sign, evaluating the scaled support sum on all 320 cosets gives 320 distinct residues. The preceding support digests and the root-vector digest bind this assertion to the exact computation; no decimal approximation enters. Distinctness modulo 692717 proves distinctness in characteristic zero, as used in [lemma 3.3](#).

A.3. The graph authority chain. The certified line-coordinate array has digest

bedb9a147cc2d196ab00b91bc147f6d544b8697fc9458ed269c0e52671a62dfd.

The reconstructed edge list and the independently generated standard line-intersection edge list have digests

c57dc120a33de30dc506e5ab045152b7d55dc2876685621ff21dad99586ffa8f,
c10e51e18748e3474a6338f612a38e28c74eee61d24922d6626aa1de04bbaa1b.

There are 135 edges, every vertex has degree 10, and the two graphs are isomorphic. This is the complement of the conventional skew-line Schlaefli graph, which is 16-regular; the two conventions have the same automorphism group. The chosen isomorphism has digest

a986674a4a0e64dc54244b81a491b86b4fe0af252fd5ad86512832641a64b0ea.

The transported standard generators have digest

e91d1484d18cca37378c5f8adbb5a74e897b0befc07a17f4cafbbbad8a99bcc1,

while the enumerated $W(E_6)$ element list has digest

2ad218397767b236bd90c78ab3a8cd8f8b791d838efd8f9be57fab9c14d1839d.

The upper-bound stabilizer cells have sizes 27, 16, 10, 6, 2, whose product is 51840. The enumerated subgroup has that same order, so the graph automorphism group and the labelled Galois image coincide exactly.

APPENDIX B. COMPLETE RESTRICTED GASSMANN AUDIT

The phrase “complete” in this appendix means complete in the fixed 350-class subgroup table, not complete among all finite groups or all arithmetic-equivalence phenomena. Each table row is represented by its full rational transitive permutation character on the 25 conjugacy classes of G . Equality of these vectors produces exactly the following eleven collision buckets:

[12, 15], [17, 21], [29, 36], [31, 39], [41, 42], [46, 48],
[57, 58], [59, 64], [112, 120], [132, 140], [301, 303].

Their indices, in the same order, are

12960, 12960, 8640, 8640, 6480, 6480, 6480, 6480, 3240, 2880, 320.

Consequently the least index in the table at which two nonconjugate subgroups have the same rational permutation character is 320, and the unique bucket at that index is [301, 303]. These are package positions used to locate embedded representatives, and are therefore locators rather than definitions.

For both H_+ and H_- , the common 25-entry vector is

(320, 0, 0, 32, 8, 20, 0, 0, 0, 0, 0, 0, 0, 2, 0, 0, 16, 0, 0, 0, 0, 4, 0, 0, 0).

The complete 350-row vector audit has digest

1738cb84d62b8011878d2af4c7929b894b24b1fb66a694c4a03266448bdc261c.

This is the object used in the proof of character equality; SmallGroup identifiers are auxiliary cross-checks only.

For clarity, the logical test applied to a candidate pair is:

B1. compare all 25 entries of $\mathbf{Q}[G/H]$;

- B2. require different conjugacy classes in the embedded group;
- B3. record subgroup order, index, core, normalizer order, and abstract identifier;
- B4. sort collision buckets by index and check that all 350 rows were visited.

The first two steps prove the Gassmann assertion, the third binds it to durable invariants, and the fourth proves the stated restricted minimum. The group-theoretic evidence file carrying these rows has digest

0b01f9d47e5141d2bff88fbe4d58ed049d88751cbf8ab1df5469009b684c4958.

The comparison with James's census is deliberately bounded. That census supplies an independent count of 350 subgroup classes, 339 rational profiles, and eleven collisions in this group [4, §3.1, p. 22]. The present audit adds a durable identification of the degree-320 bucket in one labelled line action and connects it to exact primitive resolvents and the two local branches; it does not claim the first discovery of the abstract collision.

APPENDIX C. GLOBAL RAMIFICATION LEDGER

This appendix expands the arithmetic of [section 5](#). We order the relevant subgroups as

$$(I_3, P_3, Q_3, I_5, P_5, C_3, C_2, C_\infty).$$

The common orbit-count vector on the two degree-320 carriers is

$$(36, 56, 112, 16, 64, 128, 160, 168),$$

and the corresponding local subgroup locators are

$$(140, 72, 7, 147, 23, 6, 2, 5).$$

The locators identify rows in the frozen table; the embedded permutation arrays, orbit counts, and filtration containments are the mathematical data.

The lower-filtration formula of [equation \(5.2\)](#) gives

$$\begin{aligned} v_3 &= (320 - 36) + \frac{9}{18}(320 - 56) + 6\frac{3}{18}(320 - 112) = 624, \\ v_5 &= (320 - 16) + 3\frac{5}{20}(320 - 64) = 496, \\ v_{C_3} &= 320 - 128 = 192, \\ v_{C_2} &= 320 - 160 = 160. \end{aligned}$$

This displays the lower-filtration weights responsible for every exponent; none is inferred from an expanded defining polynomial.

Let q be the certified prime

$$q = 14932047182473291995860108491583652133938007263719.$$

The exact signed identity is

$$\begin{aligned} \text{Disc}(F_\pm) &= +3^{624}5^{496}181^{192}283^{160}997^{192}1801^{160} \\ &\quad \cdot 2346241^{192}q^{160} = +3^{624}5^{496}(181 \cdot 997 \cdot 2346241)^{192}(283 \cdot 1801 \cdot q)^{160}. \end{aligned}$$

Its unsigned no-newline decimal expansion has 11,658 digits and digest

$$7f3ed0f731e5905f9af8254df2114ad15c2bb7d96cfa9a8b464a58ae8ea3ae70.$$

The decimal expansion itself is not part of the paper.

For completeness, the support proof has two inclusions. The normal closure K is unramified outside

$$\{3, 5, 181, 283, 997, 1801, 2346241, q\},$$

so its fixed fields are unramified outside this set. Conversely, after expanding the two grouped factors, every displayed prime occurs with a positive field-discriminant valuation.

Therefore every one ramifies in each $F_{\pm}$, proving equality with the displayed support. The sign is positive because the common signature is $(16, 152)$.

Finally, Artin formalism compares the full permutation representations. The exact Dedekind zeta equality

$$\zeta_{F_+}(s) = L(s, \mathbf{Q}[G/H_+]) = L(s, \mathbf{Q}[G/H_-]) = \zeta_{F_-}(s)$$

omits no bad place. We have not computed and do not claim an explicit bad Artin Euler polynomial or factor. The literal scope flag `NO_BAD_EULER_OR_ROOT_NUMBER` also excludes any decomposition Frobenius, epsilon factor, root number, or Artin-holomorphy assertion.

APPENDIX D. LOCAL BRANCH LEDGER

We give the exact one-line permutations behind the two local tables. If $[a_1, \dots, a_{27}]$ is displayed, it denotes the permutation sending i to a_i . These arrays define the local groups in the same labelled degree-27 action as the arrays defining $H_{\pm}$.

D.1. The ToM–140 branch. The decomposition group is generated by

$$\begin{aligned} D_{140} = \langle & [7, 26, 13, 12, 5, 15, 1, 18, 20, 22, 16, 4, 3, 14, 6, 11, 25, 8, 24, 9, 27, 10, 23, 19, 17, 2, 21], \\ & [17, 2, 21, 4, 18, 1, 15, 5, 20, 3, 11, 12, 22, 14, 25, 16, 6, 8, 19, 23, 10, 27, 9, 24, 7, 26, 13], \\ & [23, 24, 22, 19, 21, 20, 1, 3, 15, 13, 14, 2, 18, 16, 17, 11, 9, 10, 26, 25, 27, 8, 7, 12, 6, 4, 5] \rangle. \end{aligned}$$

Here $I_{140} = D_{140}$. The wild group P_{140} is generated by

$$\begin{aligned} & [7, 12, 8, 26, 27, 25, 23, 22, 17, 18, 16, 24, 10, 11, 9, 14, 15, 13, 4, 6, 5, 3, 1, 2, 20, 19, 21], \\ & [25, 12, 18, 26, 22, 15, 20, 13, 1, 5, 16, 24, 21, 11, 23, 14, 7, 27, 4, 17, 8, 10, 6, 2, 9, 19, 3], \end{aligned}$$

and Q_{140} is generated by

$$[6, 2, 10, 4, 8, 17, 25, 18, 23, 21, 11, 12, 27, 14, 7, 16, 1, 5, 19, 9, 3, 13, 20, 24, 15, 26, 22].$$

The containment audit certifies $Q_{140} \leq P_{140} \leq I_{140} \leq D_{140}$. Applying

$$n = [D : J], \quad e = [I : I \cap J], \quad f = \frac{|D| |I \cap J|}{|I| |J|}$$

to every double-coset representative x with $J = D \cap xH_{\pm}x^{-1}$ gives exactly the rows of [table 5](#). The representative carrier was traversed completely: both sides have 36 primes, each with $f = 1$. Thus both sides have total residue degree 36, as zeta equality requires. Their factor-degree multisets nevertheless differ because F_+ has eight degree-one factors and F_- has none.

D.2. The ToM–206 branch. The second decomposition group is generated by

$$\begin{aligned} D_{206} = \langle & [1, 2, 20, 16, 5, 18, 26, 8, 11, 23, 9, 12, 15, 22, 13, 4, 17, 6, 21, 3, 19, 14, 10, 24, 27, 7, 25], \\ & [5, 2, 3, 4, 1, 6, 14, 17, 19, 10, 21, 12, 13, 7, 15, 16, 8, 18, 9, 20, 11, 26, 23, 24, 25, 22, 27], \\ & [11, 7, 26, 24, 25, 15, 22, 14, 12, 13, 27, 21, 18, 20, 23, 4, 3, 10, 5, 8, 9, 2, 6, 16, 19, 17, 1] \rangle. \end{aligned}$$

Its inertia group is generated by

$$\begin{aligned} I_{206} = \langle & [1, 17, 22, 16, 12, 18, 26, 8, 25, 23, 27, 5, 15, 20, 13, 4, 2, 6, 21, 14, 19, 3, 10, 24, 9, 7, 11], \\ & [11, 7, 26, 24, 25, 15, 22, 14, 12, 13, 27, 21, 18, 20, 23, 4, 3, 10, 5, 8, 9, 2, 6, 16, 19, 17, 1], \\ & [12, 17, 14, 4, 1, 6, 3, 2, 19, 10, 21, 5, 13, 7, 15, 16, 8, 18, 27, 22, 25, 26, 23, 24, 11, 20, 9] \rangle. \end{aligned}$$

The group P_{206} is generated by the last two displayed generators of I_{206} , and Q_{206} by the last one. The certified containments are $Q_{206} \leq P_{206} \leq I_{206} \leq D_{206}$. The complete double-coset traversal gives [table 6](#); both sides have 18 primes, each with residue degree 2.

Their total residue degree is again 36. The separating invariant is different: F_+ has four degree-two unramified factors, whereas F_- has no factor of degree two.

D.3. Why the tables are complete. For each sign and each branch, the computation stores a canonical representative for every double coset in $D \backslash G / H_{\pm}$, forms $J = D \cap xH_{\pm}x^{-1}$, and records the quadruple $(|J|, |I \cap J|, |P \cap J|, |Q \cap J|)$. Aggregation is performed only after the sum of the double-coset sizes has been checked against $|G|$. The reported totals also satisfy

$$\sum_x e_x f_x = [G : H_{\pm}] = 320, \quad \sum_x f_x = 36,$$

on both sides and in both branches. The first identity checks the local degree; the second reconciles the residue-degree information with the ramified Euler factor. The tables therefore distinguish the products of local fields by their factor degrees without invoking any classification of the individual local extensions.

APPENDIX E. REPRODUCIBILITY LEDGER

E.1. Released input identities. The C56 and C58 paths below are relative to the released research repository; the two C59 **results/** paths are relative to the C59 project root. The two predecessor commits and their full-project manifests make the inputs retrievable independently of the present prose, while the two C59 evidence files bind the transported labels used in this paper.

TABLE 7. Released predecessor and transported-carrier identities.

Input	Exact identity	Role here
C56 line-field package	commit a55f31dca338d1e3757704b8c95d11e28c9c98d4; henon_dynamics/henon_mu3_yukawa_line_field/results/c56_certificate.json, 26739ce5aedb4a3467645f9c1b2036d4d3eec9ce4d0dbce23d67ea7b67e5fbc4; full manifest 26e3e4226cd1baea14543f14bac9ffd060ed8031741cb1ec0a38e22cd07487f4	surface, eliminant, leading coefficient, symbolic lines, and labelled $W(E_6)$ carrier
C58 ramification package	commit 184b9a8a91234a5d793d5deaa3b652ed56a524bd; henon_dynamics/henon_mu3_yukawa_line_ramification/results/c58_certificate.json, 456a481368d593f0d015436bf8a3a518d15b4567880fa7726c77d29a259d79ee; full manifest 06b7c24190d532d6c543b93f74a65d650265988734f340bbb8896953e34d3cb0	support, lower filtrations, tame classes, complex conjugation, and both D_3 alternatives
C59 resolver carrier	results/c59_resolver_evidence.json, 667e0eeb04e5724b620bf513f9556a321dfd39f9215396ed1840ca83879ec6a6	standard-label map, d_i and α_i arrays, line coordinates, split witness, and support sums
C59 group carrier	results/c59_group_evidence.json, 0b01f9d47e5141d2bff88fbe4d58ed049d88751cbf8ab1df5469009b684c4958	embedded subgroups, complete character table, filtrations, and local rows

E.2. Gate and artifact ledgers. The machine layer is organized as the eight gates in [table 8](#). The gate names, ordering, and hashes are part of the frozen contract; historical alternate numbering is not used.

TABLE 8. Canonical gate ledger.

Gate	Role	SHA-256 of gate subpayload
G0	released-authority rebind	ac445822702b5e376eed6fbfa86a4df81c7f8177ca35c8211282dca830123d5d
G1	primitive orbit resolvents	04a4272572e71467f0b50504dc1e8297248080794bbe1801961fdfea82e6a404
G2	Gassmann minimality	1738cb84d62b8011878d2af4c7929b894b24b1fb66a694c4a03266448bdc261c
G3	fixed fields and zeta	b943052f93b3f26aa19adc6ce445de9a57d957d0343b85e74f67a393cc8858b2
G4	global arithmetic	6e90a0e0f1aa451b9a0dc2e2b685e849ad6853e543744598bfb3ddcfd31b205
G5	ToM-140 local algebra	eb53f51bd0eece87d0979349faf14e791f0d8639af9c7c5d73688f14fb71d1ca
G6	ToM-206 local algebra	4dfda1903ca1f647323a1dda460818cb8596d77a3c54bd477a232188acb30121
G7	independence and scope	6b5a2aae836d3bf3392e94b915921ded235bd8a04a4c83d699c0187b2852d7be

The promoted tuple has 15 top-level payload keys and 10,412 scalar payload leaves. It consists of 13 code files and 8 result files, with 21 live scoped entries and 20 self-excluding manifest entries. The 48-test suite passed. The independent checker rebuilt the payload, rejected 20,894 one-leaf certificate mutations and eight evidence-rebind mutations, and checked exact key sets and types. These are machine-layer facts, not a claim that the present manuscript has compiled or passed a paper audit.

The principal binding values are

TABLE 9. Principal C59 binding values.

Object	SHA-256
canonical payload	a6428addfb14f00f3ed45781d9ba0944be177cfb7c257c958e7fa538fc366b
payload shape	788aa5e58d51f0d4edfa7a4e58de5748bd5a1ad1d28445d91045d5dd72c850d2
certificate	3c4c756d912d49653353503701f5b8be412d0da53383ac9c9830b6e7a953ed9a
check report	271d0123b170bef1317b63e97e3f679179b6e794185b78facd571150ba2123d3
schema	07a817bb2eade24862f0cf4dca8d1d0248eb4f473a137c07bd0200efeea8c6b4
group evidence	0b01f9d47e5141d2bfff88fbe4d58ed049d88751cbf8ab1df5469009b684c4958
resolvent evidence	667e0eeb04e5724b620bf513f9556a321dfd39f9215396ed1840ca83879ec6a6

scoped manifest	c4145ea23b57b1adcd8cfddb18c41c703e93ca8a6f84eeec b9457e0f4e046dda
-----------------	--

The finite computations use GAP 4.11.1 and TomLib 1.2.9 [8, 11]. The Python checker is a separate replay path: it consumes the promoted artifacts and reconstructs the certified projections without calling the producer’s theorem functions. The source and result manifest excludes itself, avoiding a self-hash cycle. Formal documents and this paper sit outside that historical machine tuple and therefore require their own later audits.

APPENDIX F. SOURCE AND SCOPE LEDGER

F.1. Primary-source claim matrix. Only the sources in [table 10](#) are used for mathematical context. Each use is narrower than the corresponding source.

TABLE 10. Verified source locators and their authorized roles.

Source	Locator	Role in this paper
James [4]	§3.1, printed p. 22	350 subgroup classes, 339 profiles, and eleven collisions; no present minimum-index extraction or arithmetic package.
Perlis [9]	Theorem 1, printed p. 351	classical Gassmann criteria and an explicit example; no cubic-surface construction.
Bosma–de Smit [1]	printed pp. 67–68	standard arithmetic-equivalence consequences and permutation characters.
McReynolds [7]	proof of Theorem 1.2, pp. 193–194	common-normal-closure examples with different ramified local degrees; no $W(E_6)$ realization.
Mantilla-Soler [6]	Theorem 1.3, p. 559; Theorem 3.7, p. 572	explicit equal-zeta pairs with different ramification tuples; no present pair.
Komatsu [5]	entire note	distinction between arithmetic and adelic or local equivalence.
Stauduhar [10]	abstract and opening sections	classical resolvent construction from ordered roots.
Fieker–Klüners [3]	§1, §2, Remark 2.1	relative invariants and stabilizer methods; no present split witness.
Elsenhans–Jahnel [2]	introduction theorem, p. 2; Algorithms 5.1 and A.10	explicit cubic-surface $W(E_6)$ program; no present fields.
GAP/TomLib [8, 11]	official versioned documentation	fixed-point convention and finite computation; no novelty claim and no durable field definition.

The authorized contribution remains exactly the bounded statement in [section 2](#). The matrix above records its source boundary and supplies no independent absolute-priority claim.

F.2. Thirty explicit false leaves. The promoted scope object contains exactly the following thirty Boolean false leaves. We reproduce the machine names verbatim so that prose cannot silently weaken the firewall:

```
F1. adelic_equivalence_claimed=false;
F2. artin_holomorphy_claimed=false;
F3. automorphy_claimed=false;
F4. bad_artin_euler_claimed=false;
F5. brauer_manin_claimed=false;
F6. characteristic_zero_coefficient_hash_claimed=false;
F7. class_number_equality_claimed=false;
F8. d3_branch_selected=false;
F9. decomposition_frobenius_claimed=false;
F10. expanded_characteristic_zero_resolvent_claimed=false;
F11. global_root_number_claimed=false;
F12. hasse_principle_claimed=false;
F13. hilbert_polya_operator_claimed=false;
F14. idele_group_isomorphism_claimed=false;
F15. integral_basis_claimed=false;
F16. integral_permutation_equivalence_claimed=false;
F17. local_epsilon_factor_claimed=false;
F18. local_equivalence_claimed=false;
F19. local_fields_classified_by_nefd_rows=false;
F20. local_root_number_claimed=false;
F21. maximal_order_claimed=false;
F22. monogenicity_claimed=false;
F23. motive_claimed=false;
F24. paper_complete_claimed=false;
F25. polynomial_discriminant_equals_field_discriminant_claimed=false;
F26. rational_point_claimed=false;
F27. release_claimed=false;
F28. rh_claimed=false;
F29. rings_of_integers_isomorphic_claimed=false;
F30. weak_approximation_claimed=false.
```

In addition, the literal flag

NO_BAD_EULER_OR_ROOT_NUMBER

is absolute. It forbids a decomposition-Frobenius claim at a ramified prime, a bad Artin Euler polynomial or factor, local epsilon factors, local or global root numbers, Artin holomorphy, and automorphy. The paper proves finite factor-degree separation of the two $\mathbf{Q}_3$ -algebras; it selects neither decomposition-group branch and does not classify any individual high-degree local field from a row (n, e, f, d) .

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